Shell-Coded Casimir Boundary Transduction: Physical Readout Architectures for the White Sphere–Cylinder Geometry

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Abstract

White et al. introduced a custom Casimir geometry in which a micron-scale sphere inside a conducting cylinder produced a shell-like worldline-numerics vacuum response. A role-resolved follow-on audit separated that result into morphology, calibrated scale, readout participation, and ordinary material/electromagnetic backgrounds. The durable outcome is a shell-forming finite boundary object: the annular sphere–wall region remains the physical feature to interrogate, while direct central timing is assigned to an amplitude-bounded reference role and material response becomes calibration.

This paper develops the resulting physical continuation as a boundary-transduction problem. The experimental target is a geometry-locked Casimir response associated with the annular shell role, measured through ordinary precision apparatus channels with explicit material and electromagnetic controls. The central object is a shell-coded boundary response: a finite-geometry renormalized-stress organization whose physical evidence appears as force, pressure, resonant frequency, phase, or superconducting microwave response. The metric comparison supplies scale context; the apparatus claim is a measurable boundary response.

Three readout architectures are selected by the branch triage. A paired differential-pressure cell gives the direct stress/force channel and places patch-potential, roughness, membrane, thermal, and vibration control at the center of interpretation. A high-Q electromagnetic cavity converts boundary response into a frequency, phase, or mode-overlap observable. A superconducting resonator provides a cryogenic microwave path in which shell response is decomposed alongside kinetic inductance, surface loss, vortex loading, quasiparticles, and material controls. Material impedance and capacitance measurements form the calibration channel that maps ordinary apparatus response.

The resulting program is precision finite-geometry Casimir metrology. A shell-coherent finding identifies a repeatable boundary-conditioned vacuum response in a finite shell-forming geometry and supplies a calibrated transducer for that response. Background-selected and sensitivity-bounded outcomes map how real machines turn geometry, materials, patches, resonator loss, and cryogenic conditions into measurable signals. The White sphere–cylinder system thereby becomes a benchmark for reading engineered Casimir boundaries through practical instruments.

1. INTRODUCTION

Casimir experiments translate boundary-conditioned quantum fields into measurable apparatus behavior. Conductors, dielectrics, superconductors, and structured finite geometries alter the electromagnetic mode spectrum. The renormalized stress associated with that altered spectrum appears in mechanical force, pressure, force gradients, resonant frequency shifts, phase shifts, quality-factor changes, and temperature-dependent superconducting responses. Contemporary Casimir metrology is therefore an instrumentation discipline as much as a field-theory calculation: it depends on surface potential control, separation metrology, material characterization, roughness models, finite-conductivity corrections, resonator stability, and environmental rejection [4, 5, 6, 13, 9, 11, 14, 15].

White, Vera, Han, Bruccoleri, and MacArthur used worldline numerics to study custom Casimir cavities and reported a sphere-in-cylinder case whose negative energy-density morphology resembles the shell placement drawn in an Alcubierre-style source section [1, 2]. The compact case uses a $1\,\mu$m diameter sphere centered in a nominal $4\,\mu$m diameter cylinder. The geometry separates the stress-shaping body, inner wall, annular gap, material boundary, exterior control region, and possible readout channel into distinct experimental roles.

This paper treats the White sphere–cylinder system as a finite Casimir boundary object. The program defines how the shell-coded boundary response is read by a real instrument. The required architecture is a physical measurement with a geometry-indexed response vector. A viable readout identifies a transduction channel, measures material and electromagnetic backgrounds, modulates geometry, fits a shell template against nuisance templates, and gives a physical interpretation for the recovered coefficient. Branch triage sets the route: amplitude-bounded central timing becomes a reference channel, material response becomes calibration, and pressure, cavity, and superconducting readouts become the physical continuation.

The readout program has three apparatus families. Differential pressure reads boundary stress directly through paired force cells. High-Q electromagnetic cavities read boundary response through resonant mode perturbation. Superconducting resonators read cryogenic boundary response through microwave frequency, phase, loss, and nonlinear mechanical channels. These branches are supported by existing experimental machinery: MEMS/NEMS Casimir force sensors [6, 11], high-Q nanomembrane force probes with Kelvin-probe surface-potential control [9, 10], long-range differential force measurements with explicit patch and roughness accounting [13], and superconducting Casimir or microwave-resonator platforms [12, 16, 17, 18].

2. THE PHYSICAL CLAIM

The shell-coded response is a statement about renormalized stress-energy in a finite boundary value problem. In QFT, local energy density is represented by the renormalized expectation value of the stress-energy tensor. Boundary conditions can make that local value lower than the reference vacuum in some regions while averaged constraints keep the physical system bounded. Ford's review describes this operational status of negative energy densities and the quantum inequality restrictions that accompany them [3].

Casimir apparatus makes this concept measurable. The negative sign in the ideal parallel-plate energy density expresses a boundary-conditioned stress relative to the reference state. The physical content is observed through force, pressure, energy, and mode behavior. Finite geometries add spatial structure: a boundary can concentrate the response in a specific region and the apparatus can couple to that region with varying strength.

The White sphere–cylinder geometry contributes a candidate spatial code. Its annular sphere–wall region acts as the shell role. The physical readout program tests how that role couples to a pressure cell, a cavity mode, or a superconducting resonator with enough strength and template separation for measurement against ordinary material and electromagnetic controls.

3. RELATION TO WHITE ET AL.

White et al. supplied a geometry, a worldline calculation, and a provocative metric comparison [1]. The present analysis separates the pieces needed for experimental use:

1. Geometry: a small, finite, role-structured sphere–cylinder boundary.
2. Morphology: a shell-like organization of the vacuum-boundary response.
3. Scale: the calibrated stress level associated with that response.
4. Transduction: the physical apparatus channel that converts the response into measured data.

The first two pieces specify the controlled geometry: the shell band is the region to modulate, and the body-wall spacing is the coordinate that moves the shell template. The calibrated scale piece assigns direct central timing and direct metric readout to an amplitude-bounded reference role. Material and impedance observables become calibration channels because they measure ordinary capacitance, contact, conductivity, and loss responses that share geometry controls with the shell. Differential pressure carries the direct stress measurement, with interpretation governed by patch-potential, roughness, membrane, and common-mode rejection. High-Q cavity and superconducting resonator readouts carry the resonant route: they convert boundary perturbation into frequency, phase, mode-splitting, or microwave response while retaining nuisance templates in the same fit.

3.1 Three Physical Projections of the Boundary Response

The three surviving branches are complementary projections of the same finite-boundary object. The White-derived geometry specifies a boundary-conditioned stress organization in the annular sphere–wall region. A physical instrument then chooses which projection of that object it can read: mechanical stress, electromagnetic eigenmode perturbation, or cryogenic microwave boundary response. The evidence standard is therefore branch-specific, but the source object is shared.

The pressure projection belongs to the classical Casimir measurement lineage. The ideal parallel-plate result supplies the clean stress formula, and the experimental history moved quickly into torsion balances, sphere–plane measurements, AFM probes, and MEMS/NEMS sensors because controlled micron-scale parallelism is hard [4, 5, 6, 11]. Patch potentials, roughness, finite conductivity, optical data, and electrostatic calibration are core interpretive machinery in this lineage [13, 14, 15]. In the present paper the pressure cell extends that lineage from a standard plate/sphere force law to a finite shell-coded stress contrast.

The cavity projection belongs to resonant metrology. Slater-style cavity perturbation showed that small metallic or dielectric perturbations can be read as frequency shifts whose size depends on local field overlap [7]. Modern cavity optomechanics generalizes the same instrumental idea: small boundary, material, or mechanical changes are converted into frequency, phase, linewidth, and mode-coupling observables [8]. In this branch the shell-coded boundary response is the perturbation source, and the selected cavity eigenmode is the transducer.

The superconducting projection is a cryogenic boundary-metrology problem. Superconductors alter electromagnetic boundary response, but real microwave devices are dominated by kinetic inductance, quasiparticles, vortices, dielectric loss, residual surface resistance, magnetic history, drive power, and thermal cycling [16]. The same field is now close to direct Casimir-superconductivity experiments, including condensation-energy tests, superconducting drum optomechanics, and high-resolution force sensing across a superconducting transition [12, 17, 18]. That makes the branch powerful and artifact-sensitive: small effects are amplified alongside highly structured material channels.

This projection language is the organizing principle for the apparatus. A pressure readout measures the shell geometry as differential stress. A high-Q cavity maps the same geometry onto an electromagnetic eigenmode. A superconducting resonator decomposes the geometry through cryogenic material and microwave-loss channels. The paper therefore treats the branches as field-native observables of one boundary response.

4. READOUT MODEL

Let q denote a controlled geometry or apparatus setting: gap, sphere radius or surrogate radius, cylinder radius, wall thickness, lateral offset, cell length, mode overlap, readout radius, temperature, magnetic history, or drive power. The measured observable $y(q)$ is modeled as

$$y(q) = a_{\mathrm{shell}} T_{\mathrm{shell}}(q) + \sum_i a_i T_i(q) + \varepsilon(q), \tag{1}$$

where T_{shell} is the shell-coded response template and the T_i are measured background templates. The background family includes patch potential, capacitance, roughness, membrane drift, temperature, vibration, finite conductivity, waveguide dispersion, mode mixing, kinetic inductance, surface loss, vortex response, quasiparticles, and material controls.

Equation 1 defines the evidential standard. The result is a shell coefficient with uncertainty, a set of nuisance coefficients with uncertainties, and a false-positive check from material-only and electromagnetic-only control schedules. A single absolute shift has limited meaning in this apparatus class. A geometry-locked response vector can separate shell, material, and readout behavior.

Table 1: Physical readout families for the shell-coded boundary response.

Family	Observable	Physical transduction	Main control role
Differential pressure	Paired pressure or force contrast	Boundary stress in the shell-gap geometry	Patch, roughness, membrane, thermal, and vibration rejection
High-Q cavity	$\delta f/f$, phase, Q, or mode splitting	Electromagnetic eigenmode perturbation from shell-sensitive boundary overlap	Loss, thermal, waveguide, and mode-mixing separation
Superconducting resonator	Microwave $\delta f/f$, phase, Q, nonlinear drum response, or T_{c}-coupled response	Cryogenic boundary response in a superconducting microwave apparatus	Surface loss, kinetic inductance, vortex, quasiparticle, and material separation
Material impedance	Capacitance, impedance, contact, or loss response	Ordinary material and electromagnetic apparatus behavior	Calibration map for backgrounds

The design-screening ladder associated with this paper used a signal/background recovery gate of S/B = 1.2 and an electromagnetic-only false-positive target near FP < 0.05. Table 2 summarizes the portable design implications.

5. COMMON APPARATUS ARCHITECTURE

All three readout branches require the same experimental skeleton. The apparatus contains a shell-sensitive cell, a matched dummy cell, a material-control channel, a geometry actuator, surface metrology, and a readout chain with environmental monitors. The shell cell implements the sphere–cylinder geometry or a manufacturable surrogate preserving the same shell-overlap coordinates. The dummy cell preserves material, wiring, thermal mass, and readout coupling while suppressing shell overlap. The material channel measures impedance, capacitance, contact, and loss behavior across the same schedule.

Table 2: Screening-level design constraints for the physical branches.

Branch	Representative viable cases	Predicted S/B range	Design implication
High-Q cavity	nominal clean through optimistic clean/matched/lossy	2.22–19.34	Strongest design row; mode overlap and loss templates set interpretation
Superconducting resonator	nominal balanced/clean through optimistic clean/balanced/lossy	3.59–17.76	Strong cryogenic path; surface-loss and quasiparticle controls are central
Differential pressure	optimistic balanced and optimistic quiet	2.02–5.31	Direct force path; patch and common-mode controls set feasibility
Material impedance	calibration rows	background map	Measures ordinary apparatus response for the same geometry

The common setup begins with branch-native calibration and then applies the White-geometry schedule. A pressure implementation recovers a known force or pressure law in a plate/sphere surrogate. A resonant implementation recovers a known cavity perturbation or optomechanical frequency response. A superconducting implementation records its ordinary temperature, magnetic-history, drive-power, and loss maps. Those calibrations make the shell-coded schedule evidence-bearing.

The geometry schedule is part of the measurement. Gap, radius, offset, wall thickness, length, and readout overlap move the shell template differently from patch, capacitance, material, and drift templates. Surface measurements accompany those settings. Garcia-Sanchez et al. show how Kelvin-probe surface-potential mapping can be integrated into high-Q nanomembrane Casimir measurements [9]. Bimonte et al. demonstrate a long-separation force measurement with explicit treatment of patch potentials, roughness, edge effects, uncertainty, and theory comparison [13]. Patch-specific modeling and Kelvin-probe measurements are part of the apparatus definition [14, 15].

The instrument first demonstrates ordinary Casimir sensitivity in a known geometry. This calibration step establishes separation metrology, electrostatic calibration, roughness priors, and noise floors. The shell geometry then becomes a finite-boundary test object whose output is judged by template selection and control behavior.

6. DIFFERENTIAL-PRESSURE ARCHITECTURE

Force and pressure are the original operational face of Casimir energy. The parallel-plate calculation gives the simplest stress law, and high-precision experiments largely adopted sphere–plane and microdevice geometries because large, perfectly parallel micron gaps are difficult to build and hold [4, 5, 6]. The resulting literature made surface preparation, electrostatic calibration, finite conductivity, roughness, patch potentials, and separation metrology central to the physics claim [13, 14, 15]. The White-derived pressure cell extends that lineage into a finite shell-coded stress contrast.

The differential-pressure architecture reads boundary stress directly. Two matched cells share environment and readout while their geometry settings produce different shell-gap responses. The primary observable is

$$\Delta P(q) = P_{\mathrm{shell}}(q) - P_{\mathrm{control}}(q), \tag{2}$$

with the fit model of Eq. 1. For ideal parallel plates,

$$P_{\mathrm{C}}(a) = -\frac{\pi^2 \hbar c}{240 a^4}, \tag{3}$$

which gives about 1.3×10^{-3} Pa at $a = 1\,\mu$m. The sphere–cylinder cell is finite and curved, so Eq. 3 serves as a scale reference for the apparatus model. The screening rows place the strongest pressure cases in the same metrological neighborhood, with shell pressures around 10^{-4}–10^{-3} Pa.

6.1 Expected Pressure Signatures

A White-positive pressure observation is a geometry-locked differential stress vector. The shell cell and dummy cell share environment, membrane mechanics, material stack, wiring, and readout chain. Opposite shell-gap modulation produces a pressure contrast that follows the annular sphere–wall shell template. Gap, radius, wall, offset, and length settings move the fitted shell coefficient in the White-derived schedule. Large-gap, far-field, and non-shell controls suppress that coefficient. Kelvin-probe maps and roughness measurements account for electrostatic and surface terms while leaving a residual shell-scheduled pressure component.

Alternative selections have distinct signatures. A patch-selected outcome follows measured surface-potential maps. A roughness or finite-conductivity selection follows surface and optical priors. A membrane or vibration selection follows mechanical drift monitors. An electrostatic calibration selection follows capacitance and bias schedules. A sensitivity-bounded outcome leaves the shell and nuisance coefficients below the inherited signal/background recovery gate after calibration.

6.2 Pressure Controls and Screening

The apparatus machinery follows established Casimir force metrology. Decca, Aksyuk, and Lopez review micro- and nano-electromechanical systems for Casimir measurements [6]. Stange et al. show a practical commercial MEMS force platform with piconewton-scale force resolution and electrostatic calibration [11]. Garcia-Sanchez et al. demonstrate high-Q nanomembrane force-gradient readout with in situ surface-potential control [9, 10]. These examples define the required toolchain: calibrated actuation, separation sensing, patch mapping, roughness measurement, and common-mode rejection.

In the priority physical screen, pressure survives in the two strong-control optimistic rows: S/B = 5.31 for the quiet case and S/B = 2.02 for the balanced case. The branch therefore asks for a paired-cell Lifshitz pressure model, patch-potential map prior, roughness and membrane-stress model, and validated common-mode rejection.

7. HIGH-Q CAVITY ARCHITECTURE

The high-Q cavity branch belongs to the cavity perturbation and resonant metrology tradition. In a resonator, a small boundary, dielectric, conductor, or mechanical perturbation becomes a shift in frequency, phase, linewidth, or mode splitting. Maier and Slater's resonant-cavity perturbation work already used small inserted objects to infer local electromagnetic field components from frequency shifts [7]. Cavity optomechanics later made the same transduction logic a broad precision-measurement platform in which mechanical displacement and boundary motion modulate optical or microwave eigenmodes [8]. The White shell geometry therefore enters this branch as a boundary perturbation source whose usefulness is determined by mode overlap.

The high-Q cavity architecture uses resonance as the transducer. A cavity mode with large field overlap in the shell-sensitive region converts a small boundary perturbation into a measurable frequency, phase, Q, or mode-splitting signal. In first-order perturbation language,

$$\frac{\delta f}{f} \sim -\frac{1}{2}\frac{\int_\Omega \left(\delta\epsilon\,|E|^2 - \delta\mu\,|H|^2\right)\,dV}{\int_\Omega \left(\epsilon|E|^2 + \mu|H|^2\right)\,dV} + \Delta_{\mathrm{boundary}}, \tag{4}$$

where $\Delta_{\mathrm{boundary}}$ denotes the shell-sensitive finite boundary contribution modeled by the apparatus eigenmode solver. The equation states the design requirement: maximize known shell overlap and measure the ordinary material terms in the same basis. The central theoretical variables are shell-mode overlap and nuisance separation: the shell-coded boundary response projects onto a chosen eigenmode with a schedule distinct from thermal, loss, waveguide, and mode-mixing channels in a successful readout.

7.1 Expected Cavity Signatures

A White-positive cavity observation is a mode-indexed shell coefficient. A high-shell-overlap mode shows a frequency, phase, Q, or mode-splitting response that follows the White shell schedule. A low-shell-overlap mode, null mode, or dummy cavity suppresses the coefficient. The eigenmode solver predicts the sign and relative magnitude trend across gap, radius, wall, offset, and mode-overlap settings. Thermal expansion, conductor loss, waveguide, port coupling, and material-wall templates remain separately fitted coordinates.

Alternative selections have branch-specific signatures. A mode-mixing selection follows avoided crossings or mode-family exchange. A thermal selection follows expansion monitors. A conductor-loss or finite-conductivity selection follows surface and optical priors. A waveguide or port-coupling selection follows readout-chain sweeps. A material-wall selection follows the dummy cavity or material-control channel. A sensitivity-bounded outcome indicates insufficient shell overlap or excess resonator noise in the tested mode.

The screening ladder ranks this branch highly. The strongest cavity row gives S/B = 19.34, and the nominal clean row remains above the recovery gate. Those values require measured templates for conductor loss, thermal expansion, waveguide dispersion, port coupling, and mode mixing. A strong cavity design therefore includes a shell-overlap mode, a low-overlap or null mode, a dummy cavity, independent thermal metrology, deliberate port-coupling sweeps, and a validated eigenmode model.

8. SUPERCONDUCTING-RESONATOR ARCHITECTURE

The superconducting branch sits at the intersection of superconducting microwave resonators, Casimir forces involving superconductors, condensation energy, and cryogenic optomechanics. Superconducting boundaries change the electromagnetic response of a cavity, but practical resonator behavior is often set by kinetic inductance, quasiparticles, trapped vortices, dielectric participation, residual surface resistance, drive-power dependence, and thermal history [16]. Recent Casimir-superconductivity experiments make the branch experimentally timely, from NEMS transition-temperature concepts to superconducting drum force readout and on-chip force sensing across a superconducting transition [12, 17, 18]. The branch is therefore read as superconducting-boundary metrology: the shell-coded Casimir geometry is tested inside the loss-channel decomposition of a real cryogenic microwave device.

The superconducting architecture uses a cryogenic microwave resonator, drum, or a cavity sensitive to the superconducting transition to read the shell-coded response. The observable can be frequency shift, phase shift, Q change, nonlinear mechanical response, or a superconducting transition response. The core fit is

$$\left(\frac{\delta f}{f}\right)_{\mathrm{sc}} = a_{\mathrm{shell}}T_{\mathrm{shell}} + a_{\mathrm{ki}}T_{\mathrm{ki}} + a_{\mathrm{surf}}T_{\mathrm{surf}} + a_{\mathrm{vort}}T_{\mathrm{vort}} + a_{\mathrm{qp}}T_{\mathrm{qp}} + a_{\mathrm{mat}}T_{\mathrm{mat}} + \varepsilon, \tag{5}$$

with kinetic inductance, surface loss, vortex loading, quasiparticles, and material response treated as measured templates.

8.1 Expected Superconducting Signatures

A White-positive superconducting observation is a microwave response locked to the shell schedule after cryogenic material terms are measured. A shell-coupled resonator differs from a material-only resonator on the same cooldown path. Low-shell-overlap modes, dummy resonators, or non-shell geometries suppress the coefficient. Temperature sweeps below and across T_c, magnetic-history tests, vortex-loading controls, drive-power sweeps, and quasiparticle checks leave a stable shell coefficient whose geometry dependence follows the sphere–wall schedule.

Alternative selections are branch-defining. A kinetic-inductance selection follows temperature, current, and film participation. A quasiparticle selection follows nonequilibrium population and drive history. A vortex selection follows magnetic preparation and cooldown history. A surface-loss or dielectric-loss selection follows Q, field participation, and power sweeps. A material-only selection appears in the control resonator with the same schedule. A sensitivity-bounded or nuisance-degenerate outcome leaves the shell coefficient below the recovery gate or too close to a measured loss-channel template.

This branch has unusually strong source support. Perez et al. designed a NEMS system to probe Casimir energy corrections to superconducting condensation energy by measuring a superconducting film in a controlled cavity [12]. de Jong et al. report a superconducting Casimir force measurement through nonlinear dynamics of a superconducting drum resonator in a microwave optomechanical system [17]. Xu et al. add a parallel cryogenic plate platform aimed at Casimir force sensing across a superconducting transition [18]. Gurevich reviews the microwave-loss mechanisms that govern superconducting resonators, including quasiparticles, residual surface resistance, dielectric loss, trapped vortices, kinetic inductance effects, and high-field limits [16]. Those mechanisms are the exact nuisance family in Eq. 5.

The screening ladder gives superconducting rows from S/B = 3.59 to S/B = 17.76 across the viable nominal and optimistic cases. The apparatus includes shell-coupled and material-only resonators on the same cooldown path. Temperature sweeps across and below T_c, magnetic-history tests, vortex-loading controls, drive-power sweeps, quasiparticle checks, and geometry modulation all become evidence-bearing coordinates.

9. CONTROL MATRIX AND OUTCOME CLASSES

The control channels are measured observables. They identify where the White-derived geometry schedule lands: shell response, material response, resonator response, sensitivity bound, or central-reference channel. Table 3 summarizes the control standard for each physical branch.

Table 3: Evidence-bearing controls for physical readout branches.

Branch	Required controls	Shell-coherent data signature
Differential pressure	Matched dummy cell, opposite gap schedule, Kelvin-probe map, roughness prior, membrane drift model, thermal and vibration monitors	Pressure vector selects the shell schedule after patch, roughness, and membrane templates are included
High-Q cavity	Eigenmode model, shell-overlap and low-overlap modes, dummy cavity, thermal expansion monitor, loss and port-coupling sweeps	Frequency or phase vector selects shell overlap over loss, thermal, waveguide, and mode-mixing templates
Superconducting resonator	Material-only resonator, temperature sweep, drive-power sweep, magnetic-history control, quasiparticle checks, surface-loss model	Microwave response selects shell schedule after kinetic-inductance, loss, vortex, quasiparticle, and material templates are included

Table 4: Outcome classes for shell-geometry measurements.

Outcome	Data signature	Physical meaning
Shell coherent	Shell coefficient exceeds recovery gate and controls satisfy the false-positive target	The White sphere–cylinder shell schedule appears as a measurable Casimir/Lifshitz boundary response in that transducer
Material selected	Patch, capacitance, impedance, contact, or loss template has the stable coefficient	Ordinary material or electromagnetic response carries or masks the White shell schedule
Resonator selected	Cavity or superconducting response follows mode mixing, conductor loss, surface loss, vortex, or quasiparticle template	Resonator material dynamics explain the observed modulation under the tested settings
Sensitivity bounded	Shell and nuisance templates remain below recovery gate after calibration	The branch sets an upper bound on transducible White-shell response
Central reference bounded	Central propagation/timing coefficient remains far below the recovery gate	White-style central transit readout remains below laboratory recovery scale in this audit class

10. IMPLICATIONS FOR THE WHITE SPHERE–CYLINDER OBSERVATION

A shell-coherent pressure result means the White annular sphere–wall morphology has a direct mechanical stress projection. The numerical shell-shaped energy-density pattern becomes a measured finite-geometry Casimir/Lifshitz stress contrast with the same shell schedule. The interpretation is a finite-boundary stress result tied to the White geometry.

A shell-coherent high-Q cavity result means the White shell response projects onto an electromagnetic eigenmode. The White geometry becomes a resonant boundary perturbation whose coefficient appears in frequency, phase, linewidth, or mode splitting after loss and thermal controls are included. The interpretation is an eigenmode readout of the finite boundary geometry.

A shell-coherent superconducting resonator result means the White shell-coded boundary response survives cryogenic microwave loss-channel decomposition. The same annular shell schedule appears after kinetic-inductance, surface-loss, vortex, quasiparticle, magnetic-history, and material controls are measured. The interpretation is a superconducting-boundary projection of the White finite Casimir geometry.

A material-, patch-, or loss-selected result means the White geometry schedule is experimentally carried by ordinary material or electromagnetic response in that apparatus. The numerical shell morphology remains a finite-boundary calculation, while the laboratory readout assigns the measured coefficient to surface potential, capacitance, contact, conductivity, thermal, waveguide, or microwave-loss physics.

A sensitivity-bounded result places the White-shell coefficient below the required signal/background gate in the tested branch. The result is an upper bound on measurable White-shell transduction for that physical projection. The central timing reference has the same interpretive role: the White-style current, photon, or electron transit channel remains below the laboratory recovery scale under the modeled backgrounds.

Across all branches, the physics is finite-boundary QFT metrology. The measured quantity is the consequence of boundary-conditioned renormalized stress as read through a calibrated apparatus channel. The metric comparison remains a scale reference and a motivation for the shell geometry.

11. OUTCOME-DEPENDENT WHITE-GEOMETRY INTERPRETATIONS

The outcome classes interpret White's sphere-cylinder observation. The primary question is where the White shell coefficient lands after branch-native controls.

If all three shell branches select the shell coefficient, the White sphere–cylinder geometry has mechanical, electromagnetic eigenmode, and cryogenic microwave projections. That cross-branch pattern establishes the annular sphere–wall region as a measurable finite-boundary response object.

If pressure selects the shell coefficient while resonant branches remain bounded, the White shell response is strongest as a direct stress projection in the tested designs. The resonant nulls then constrain mode overlap and microwave coupling while the pressure branch carries the mechanical projection.

If the high-Q cavity selects the shell coefficient while pressure remains bounded, the White shell response is more visible as an electromagnetic eigenmode perturbation than as a direct force contrast in the tested pressure cell. The result points to mode-overlap engineering as the active route.

If the superconducting branch alone selects the shell coefficient, the White schedule has a cryogenic microwave projection with high artifact sensitivity. That pattern assigns special weight to material-only resonators, magnetic-history controls, temperature sweeps, and drive-power sweeps.

If material, patch, or resonator-loss templates select the stable coefficient, the White geometry schedule lands in an ordinary apparatus channel in that branch. Those outcomes interpret the White-style schedule as background-coupled in that implementation and define the controls for a later shell-coherence test.

If every branch is sensitivity bounded, the current readout line closes at the tested sensitivity. The result becomes a quantitative upper-bound statement on pressure, cavity, and superconducting transduction of the White shell coefficient.

12. REPRODUCIBILITY AND SOURCE BASIS

The citable scientific basis for this manuscript is the public literature in the reference list. The references provide the canonical DOI, arXiv, NIST, and journal locations for the source material.

The numerical screening values quoted in Tables 2 and 3 come from a finite readout-ladder calculation using shell-template recovery, electromagnetic-only false-positive checks, and physical backend cases for pressure, cavity, superconducting, material, and central-reference channels. The associated supplementary material consists of compact machine-readable screening summaries: candidate definitions, backend physical cases, gate summaries, and run-level summary metadata. Archiving the manuscript source and PDF with those summaries keeps the quoted design-screening values traceable.

13. CONCLUSION

The White sphere–cylinder geometry has an experimental role map. The sphere and inner cylinder wall set the finite boundary condition. The annular gap carries the shell template. The material wall, exterior region, and central channel supply control and reference roles. The prior audit maps those roles onto apparatus choices: shell morphology motivates modulation of the sphere–wall gap, calibrated scale makes central timing an amplitude-bounded reference, material and impedance response become calibration, and the remaining physical question is transduction.

Pressure, cavity, and superconducting readouts are the three branch-native projections of that transduction problem. A pressure apparatus recovers or bounds a shell-indexed stress contrast after patch maps, roughness priors, membrane drift, and common-mode subtraction are included. A high-Q cavity recovers or bounds the same shell coordinate as an electromagnetic eigenmode perturbation after conductor loss, thermal expansion, waveguide coupling, and mode mixing are measured. A superconducting resonator recovers or bounds the shell coefficient after kinetic inductance, surface loss, vortex loading, quasiparticles, and material controls are included in the same fit.

The outcome classes give direct interpretations for White's observation. Shell coherence assigns the annular sphere–wall morphology to a measurable finite-boundary response in a specific physical projection. Material or resonator selection assigns the same geometry schedule to ordinary apparatus physics. Sensitivity bounds set upper limits on measurable White-shell transduction in the tested branch. Central timing remains an amplitude-bounded reference channel in the current audit.

The program converts the original visual metric comparison into a field-native boundary-readout problem. The next physical result identifies where the White shell coefficient lands: direct pressure, electromagnetic mode shift, superconducting microwave response, material/background response, or an upper bound. That answer sharpens how engineered vacuum-boundary structure becomes measurable force, pressure, frequency, phase, or cryogenic resonator response.

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